UNIVERSITY OF TOMORROW

SCHOOL OF COMPUTER SCIENCE UPES, BIDHOLI,

DEHRADUN, UTTRAKHAND, INDIA

MINOR PROJECT

SYNOPSIS REPORT ON:

SPG 6/2/25

DAMAGE OBJECT DETECTION IN A MOVING CONVEYOR BELT

UNDER THE GUIDANCE OF DR. SUBHRANIL DAS

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Synopsis is done.
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Abstract:

In modern manufacturing, the detection of damaged objects on a moving conveyor belt is crucial for ensuring product quality and operational efficiency.

This paper investigates the use of advanced technologies such as computer vision, machine learning, and sensor-based systems for automatic damage detection.

By leveraging high-resolution cameras, image processing algorithms, and deep learning models, the system identifies and classifies defective objects in real-time, enabling the removal of damaged items before they proceed further in production.

Challenges such as motion blur, lighting variations, and object overlap are addressed through optimized detection techniques. The study also explores future directions such as the integration of edge AI and IoT for enhanced accuracy and adaptability.

Introduction:

Manufacturing processes are rapidly advancing towards full automation, driven by the need for higher efficiency, lower production costs, and improved quality assurance. One of the most critical aspects of modern manufacturing is quality control, which ensures that each product leaving the production line meets stringent specifications and quality standards. Defective products can arise from various factors, such as improper handling, equipment malfunction, or material defects.

Damage incurred during production can compromise the integrity of the final product, leading to negative consequences such as financial losses due to scrapped goods, product recalls, customer dissatisfaction, and even potential harm to brand reputation.

Traditionally, quality control in manufacturing environments has relied heavily on manual inspection, where human operators visually assess products for defects or damage. While this approach has been effective in smaller-scale or low-speed production environments, it has significant limitations in modern high-speed manufacturing lines.

Manual inspections are not only labor-intensive but also subject to human error, fatigue, and inconsistency. As production speeds increase, human inspectors struggle to maintain accuracy and attention to detail, leading to missed defects, undetected damage, and inconsistent product quality. In industries that demand precision, such as electronics, automotive, or pharmaceuticals, even a small defect can have significant repercussions.

PROBLEM STATEMENT:

In high-speed manufacturing environments, damaged objects frequently pass undetected through production lines, leading to defects in the final product. Manual inspection is not feasible for large-scale, high-speed production due to human limitations and inconsistencies. Therefore, there is a need for a robust, automated system capable of detecting various types of damage (e.g., cracks, deformations, or foreign particles) on a moving conveyor belt. The problem is further complicated by the presence of motion blur, inconsistent lighting conditions, and overlapping objects, which make detection challenging.

LITERATURE REVIEW:

Automated damage detection has been a focus of research for several decades, with various approaches explored to solve the problem. Computer vision has been widely used in the past, primarily through edge detection and pattern recognition techniques (Huang et al., 2016). Recent advancements in machine learning, especially in the development of Convolutional Neural Networks (CNNs), have greatly improved the accuracy of damage detection in complex environments (Zhou et al., 2019).

Infrared sensors have been utilized to detect internal defects, particularly in materials such as plastic and ceramics. The use of deep learning for anomaly detection has also gained traction, with unsupervised learning models offering potential solutions to detect previously unseen defects.

Despite these advancements, many existing systems struggle with real-time detection and accuracy in variable environmental conditions, necessitating further research.

OBJECTIVES:

The primary objectives of this study are as follows:

1. Develop a real-time, automated system for detecting damaged objects on a moving conveyor belt using computer vision and machine learning techniques.
2. Address challenges such as motion blur, lighting variations, and overlapping objects through optimized algorithms and hardware configurations.

METHODOLOGY:

The methodology for this study involves several key steps:

1. System Setup: High-resolution cameras and sensors will be positioned along the conveyor belt to capture continuous image and sensor data of the objects. The system will be designed to handle various object types and surface textures.

2. Data Collection: A dataset will be created or we use preexixting data consisting of images of both damaged and undamaged objects, capturing various types of defects such as cracks, deformities, and scratches. Additional data may be gathered from sensors such as infrared or X-ray for detecting internal damage.

3. Preprocessing: Images and sensor data will be preprocessed to remove noise and standardize the inputs. Techniques such as histogram equalization, normalization, and motion compensation will be used to account for lighting variations and motion blur.

4. Model Development: Machine learning models, specifically Convolutional Neural Networks (CNNs), will be trained using the labeled dataset. The model will be designed to classify objects into "damaged" or "undamaged" categories, with a focus on minimizing false positives and false negatives.

5. Real-Time Processing: The trained model will be deployed for real-time damage detection. Edge computing solutions may be employed to ensure low- latency processing. Algorithms will be optimized for handling overlapping objects and detecting defects under varied lighting conditions.

6. Automation Response: Once a damaged object is detected, the system will trigger an automated actuator to remove the defective item from the conveyor belt. This could be achieved through robotic arms or pneumatic ejectors.

7. Evaluation: The system will be tested in a real manufacturing environment. Performance metrics such as detection accuracy, processing speed, and false positive rates will be measured and analyzed to evaluate the effectiveness of the approach.

8. Future Enhancements: The potential for integrating IoT devices and edge AI to improve the system's adaptability and scalability will be explored as part of future work.

ROADMAP:

This condensed roadmap outlines a 5-month timeline for the development and deployment of an automated damage detection system in a manufacturing environment. The focus is on rapid prototyping, integration, and testing to deliver a functional solution within a shorter time frame.

PHASE 1: RESEARCH & PLANNING (MONTH 1)

- **OBJECTIVE**: Quickly define the project scope, gather requirements, and evaluate potential technologies.
  - **MILESTONES:**
    - Perform a rapid review of existing damage detection systems and technologies.
    - Identify the key types of damage (e.g., cracks, deformities) that the system will detect.
    - Finalize hardware and software requirements, including cameras, sensors, and computing platforms.
    - Define project objectives, success criteria, and a fast-tracked timeline for development.

PHASE 2: SYSTEM DESIGN & HARDWARE SETUP (MONTH 1)

- **OBJECTIVE:** Design the system architecture and install the necessary hardware for testing.
  - **MILESTONES:**
    - Design the basic system architecture, including camera placement, sensor integration, and data pipelines.
    - Procure high-resolution cameras, sensors (optional if time permits), and computing hardware.
    - Set up the test environment with the conveyor belt and configure lighting conditions for optimal image capture.
    - Begin data acquisition by capturing images of objects moving on the belt under controlled conditions.

PHASE 3: DATA COLLECTION & PREPROCESSING (MONTH 2)

- **OBJECTIVE:** Collect and preprocess data quickly for initial model development.
  - **MILESTONES:**
    - Capture images and sensor data (if available) of damaged and undamaged objects.
    - Label and categorize the dataset with a focus on the most common damage types.
    - Implement basic preprocessing techniques (e.g., noise reduction, normalization) to prepare data for training.
    - Augment data to increase variability and robustness of the model (e.g., rotation, scaling).

PHASE 4: MODEL DEVELOPMENT (MONTH 2-3)

- **OBJECTIVE:** Develop and train machine learning models for damage detection using available data.
  - **MILESTONES:**
    - Develop a simple Convolutional Neural Network (CNN) for image-based damage detection.
    - Train the model with the preprocessed dataset and test it using a validation set.
    - Evaluate initial model performance in terms of accuracy and speed, and make quick adjustments to improve performance.
    - Begin exploring methods to handle motion blur and lighting variations through data augmentation and model tuning.

PHASE 5: REAL-TIME PROCESSING INTEGRATION (MONTH 2-3)

- **OBJECTIVE:** Integrate the model with real-time processing capabilities on the conveyor belt system.
  - **MILESTONES:**
    - Deploy the trained model into a real-time processing system using edge computing where applicable.

        - Integrate the camera feed and model output with the conveyor belt control system.
        - Test real-time detection accuracy on moving objects, adjusting for conveyor speed and object variability.
        - Implement basic algorithms to handle overlapping objects and detect under variable lighting conditions.

PHASE 6: AUTOMATION & ACTUATION (MONTH 3)

- **OBJECTIVE:** Implement automated object removal based on detection results.
    - **MILESTONES:**
        - Connect the detection system with an actuator (e.g., robotic arm or pneumatic ejector) to remove damaged objects from the conveyor belt.
        - Test the actuator's response time and accuracy in removing defective items from the production line.
        - Refine the integration between detection and actuation systems for smooth and reliable operation.

PHASE 7: TESTING & REFINEMENT (MONTH 3-4)

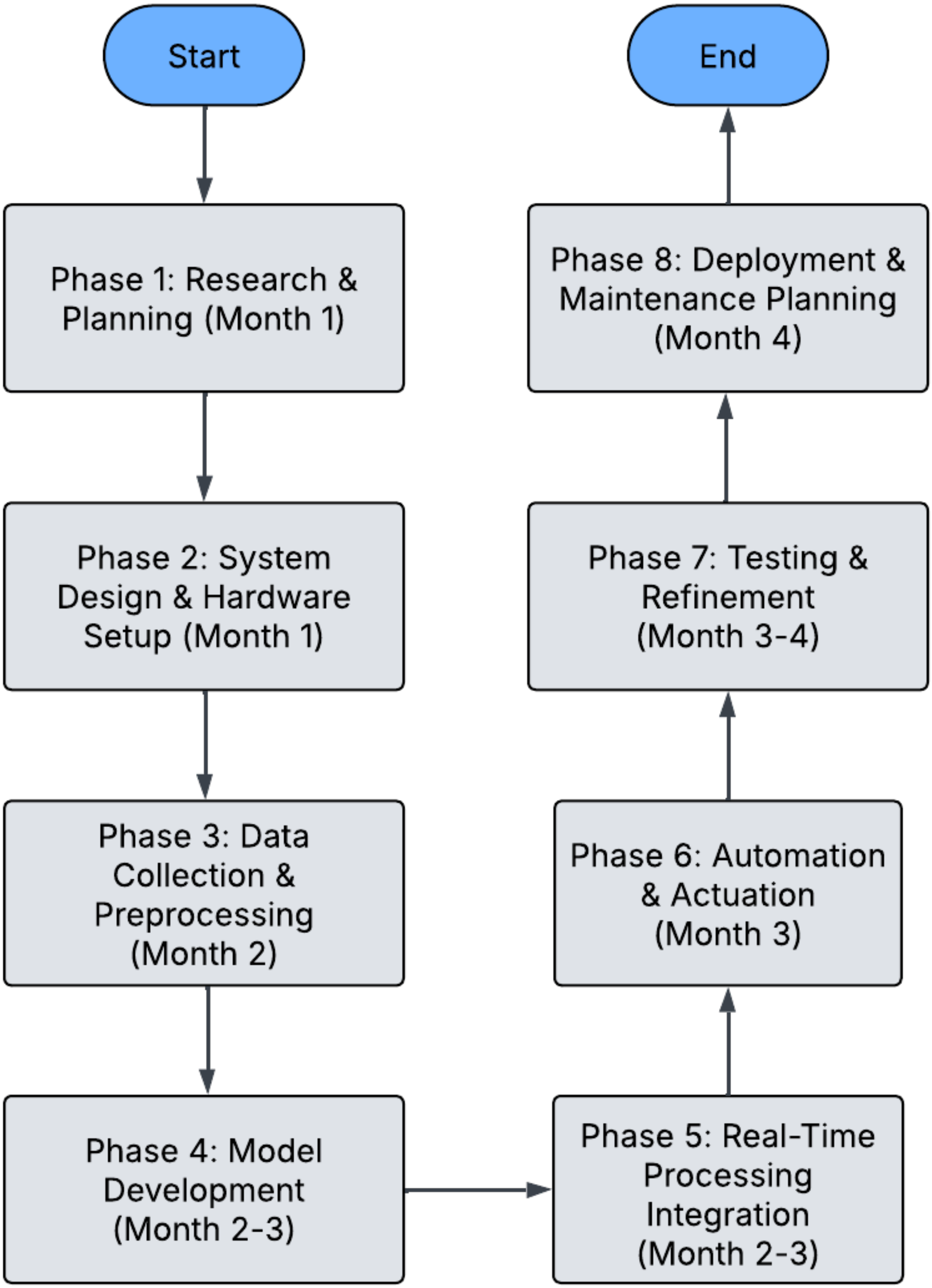
- **OBJECTIVE:** Conduct final testing and refine the system for deployment.
    - **MILESTONES:**
        - Perform extensive testing in a real manufacturing environment with various object types and damage scenarios.
        - Collect performance metrics such as detection accuracy, false positive/negative rates, and processing speed.
        - Make necessary refinements to the model, detection algorithms, and actuation response to ensure reliability.
        - Ensure the system meets operational standards for deployment in the manufacturing line.

PHASE 8: DEPLOYMENT & MAINTENANCE PLANNING (MONTH 4)

- **OBJECTIVE:** Finalize the system for deployment and plan for maintenance.

- **MILESTONES:**
  - Deploy the damage detection system into the production environment.
  - Train staff on system operation and routine maintenance procedures.
  - Establish a plan for regular system updates and improvements based on ongoing data collection.
  - Explore the possibility of future enhancements, such as integrating IoT devices or implementing edge AI for improved scalability.

PERT CHART:


Start
Phase 1: Research & Planning (Month 1)
Phase 2: System Design & Hardware Setup (Month 1)
Phase 3: Data Collection & Preprocessing (Month 2)
Phase 4: Model Development (Month 2-3)
Phase 5: Real-Time Processing Integration (Month 2-3)
Phase 6: Automation & Actuation (Month 3)
Phase 7: Testing & Refinement (Month 3-4)
Phase 8: Deployment & Maintenance Planning (Month 4)
End

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